

Module 02: Agents — Autopoiesis, Agency, and the Self-Organizing System

Learning Objectives

1. Explain how the concept of **agency** emerges from the Markov Blanket formalism when active states are distinguished from sensory states.
2. Compare autopoietic, enactivist, and Active Inference accounts of what it means to be an agent.
3. Critically assess whether agency is an intrinsic property of systems or a description imposed by observers.

Introduction

In Module 01, we established that systems are defined by their boundaries. But not all bounded systems are *agents*. A rock has a boundary, yet we do not attribute agency to it. What additional feature transforms a system into an agent? In Active Inference, the answer lies in the asymmetry of the Markov Blanket: an agent is a system whose active states influence external states in a way that is conditioned on — and serves to maintain — its internal states.

This module explores the philosophical depth of this claim, connecting it to classical discussions of agency in phenomenology (Husserl's intentionality), pragmatism (Dewey's organism-environment transaction), and enactivism (Varela's sense-making).

Key Concepts

1. From Boundary to Agency

A Markov Blanket has two faces: sensory states (how the world affects the system) and active states (how the system affects the world). An agent arises when active states are not random but serve to minimize the system's free energy — that is, when the system acts *so as to*

maintain itself. Agency, on this account, is not a mysterious faculty but a consequence of self-organization.

Philosophically, this resonates with Spinoza's *conatus* — the striving of every being to persist in its own existence — and with Heidegger's concept of *Sorge* (care): to be an agent is to be concerned with one's own continuation.

2. Self-Models and Minimal Selfhood

Does an agent need a *self*? The philosopher Thomas Metzinger argues that selfhood requires a self-model — an internal representation of oneself as an entity distinct from the world. In Active Inference terms, this corresponds to the agent's generative model containing a representation of its own Markov Blanket. Even a bacterium, insofar as its metabolic network "models" the boundary conditions for its own persistence, instantiates a minimal form of selfhood.

This raises the question: is there a continuum from bacterial self-organization to human self-consciousness? And if so, where do morally relevant properties like sentience and autonomy enter?

3. The Problem of Other Agents

How do we recognize other agents? The "problem of other minds" — how I can know that you are conscious — takes new form in Active Inference. If I model you as a system with active states that minimize your free energy, I am attributing agency to you. This attribution is itself an inference: I infer your internal states through the lens of my own generative model.

This connects to Husserl's concept of *Einfühlung* (empathy) and to the simulation theory / theory-theory debate in philosophy of mind.

Applications

- **Artificial Agency:** If agency is defined by self-organizing dynamics rather than biological substrate, do thermostat-like systems count as agents? What about reinforcement learning algorithms?

- **Moral Status:** If agency is graded rather than binary, what implications does this have for the moral status of animals, ecosystems, and AI systems?

Conclusion

Agency is not a primitive concept but an emergent one: it arises when a Markov Blanket system acts to maintain itself. This reframes classical philosophical debates about free will, intentionality, and moral status. In Module 03, we turn to the agent's most fundamental activity — perception.